WANDA ESTU
KURNIAWAN

DIGITAL SPACE GOVERNANCE

BY COMBATING

"CYBERBULLYING"



5 CLASSIC MODEL VS RoBERTa

2-STAGES CLASSIFICATION

BACKGROUND

High Cyberbullying Rates

Cyberbullying cases on social media continue to rise and severely damage victims' mental health.

Characteristics of Twitter

The rapid flow of information makes Twitter prone to becoming a breeding ground for hate speech.

Limitations of Manual Monitoring

Reviewing millions of tweets manually is inefficient and unsustainable for moderator teams.

Machine Learning Automation Solution

Natural Language Processing (NLP) technology enables deep contextual understanding and fast, accurate, automated cyberbullying detection.



PROJECT OBJECTIVES

IN COMBATING *CYBERBULLYING*

AUTOMATED DETECTION

Build a system capable of quickly and accurately identifying cyberbullying-related tweets.

PREVENT SPREAD

Enable early intervention before bullying content goes viral and spreads widely.

CREATING SAFE SPACES

Reduce cybercrime rates and support users' mental health on the Twitter platform.

CORE GOVERNANCE PILLARS

IN COMBATING CYBERBULLYING

Collection, cleaning, labeling, and privacy protection of Twitter tweet data.

DATA GOVERNANCE

MODEL GOVERNANCE

Training, accuracy evaluation, and continuous updating of NLP algorithms to stay aligned with language evolution.

INFRASTRUCTURE GOVERNANCE

Integration pipeline setup, deployment, and real-time system performance monitoring.

Algorithmic bias mitigation, transparency in bullying criteria, and adherence to platform policies and applicable laws.

COMPLIANCE & LEGAL GOVERNANC

TARGET : TWEET TEXT



DATASET DESCRIPTION

This dataset comprises thousands of tweets collected from Twitter (X) for cyberbullying analysis and classification. It is used to train Machine Learning / Natural Language Processing (NLP) models to detect and categorize cyberbullying content into specific categories or severity levels.

DATA VOLUME

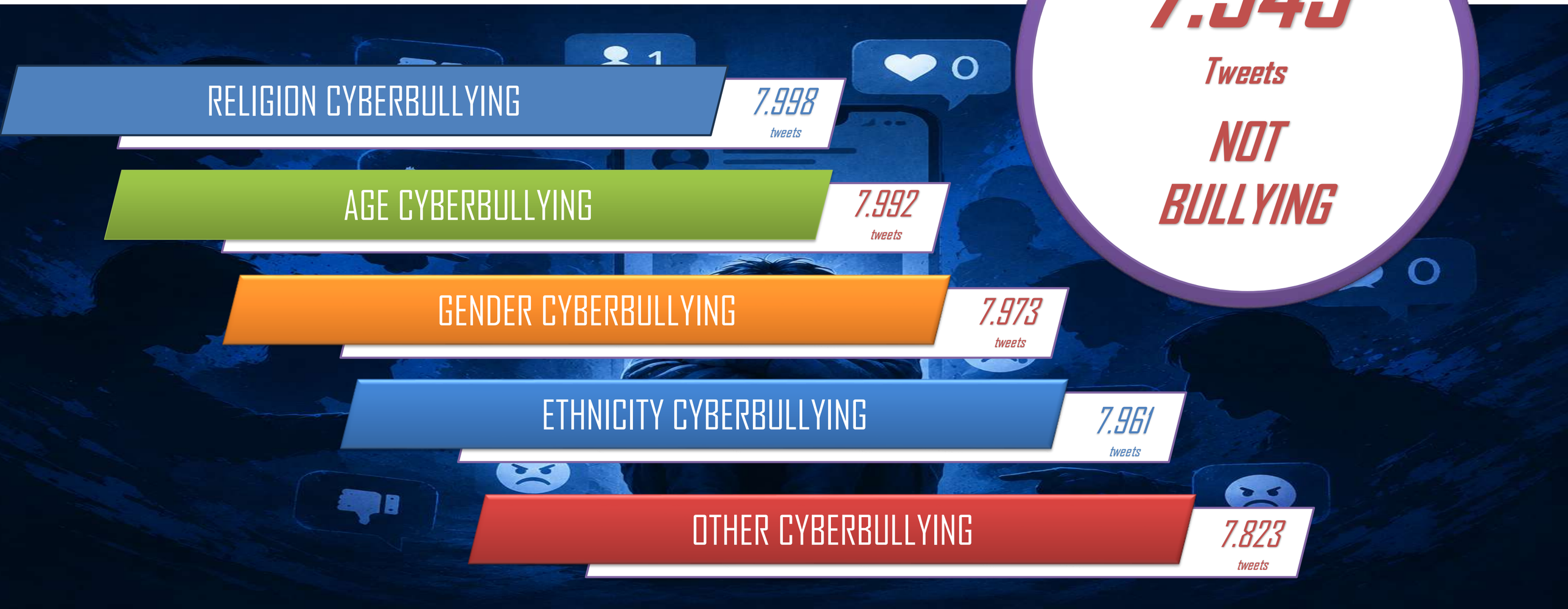
47.692

TWEETS

SOURCE OF DATASET : KAGGLE

DATASET TITLE :
TWITTER CYBERBULLYING DATASET

CYBERBULLYING CATEGORY



EKSPLORASI JENIS *CYBERBULLYING*

WORD CLOUD DAN HORIZONTAL BAR CHART





5 TOP KEYWORDS





GENDER BULLYING

Focuses on gender-based harassment. Dominated by harassment/sexuality issues (*rape, gay, sexist*) and gender-targeted slurs (*women, bitch*).

EKSPLORASI JENIS *CYBERBULLYING*

WORD CLOUD DAN HORIZONTAL BAR CHART





5 TOP KEYWORDS



AGE BULLYING

Focuses on school and youth bullying. Dominated by educational levels (*high school, middle school*) and target identifiers (*girl, kid, boy*).

EKSPLORASI JENIS *CYBERBULLYING*

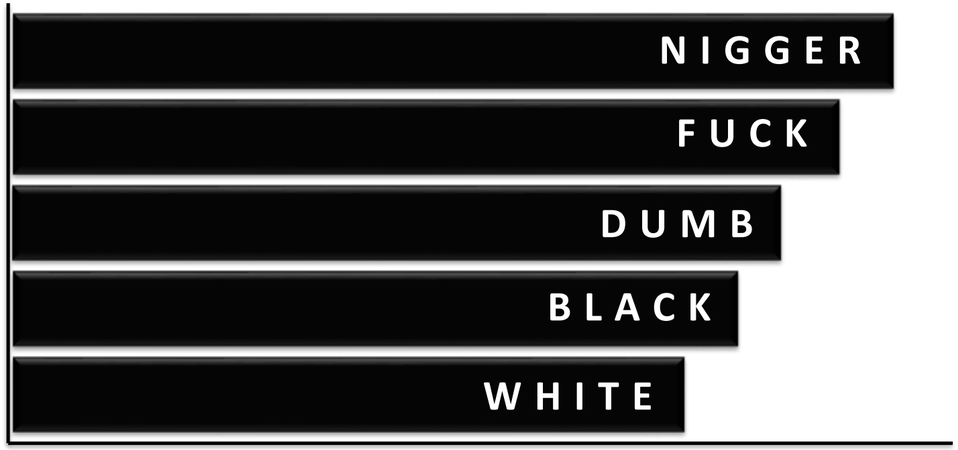
WORD CLOUD DAN HORIZONTAL BAR CHART





5 TOP KEYWORDS





ETHNICITY BULLYING

Focuses on racism and race/ethnicity-based harassment. Highly sensitive, dominated by extreme racial slurs (*nigger*), profanity (*fuck, dumb*), and racial descriptors (*black, white*).

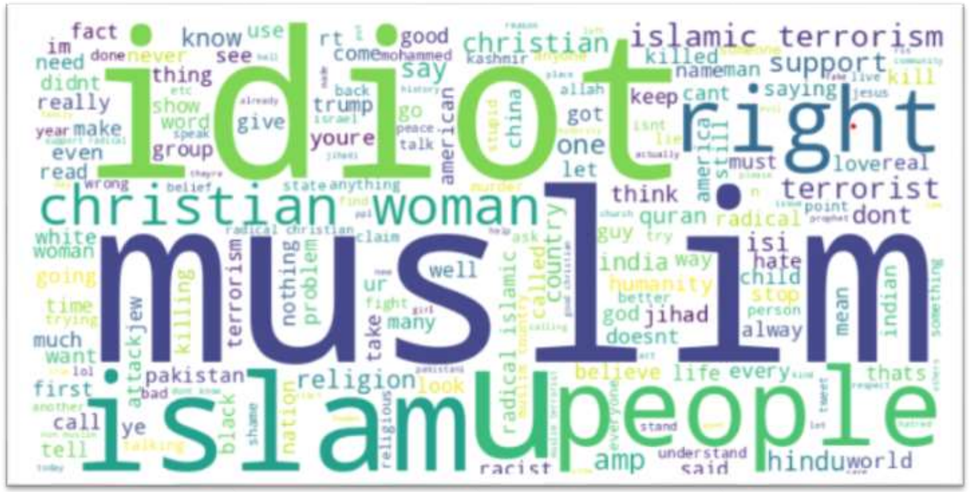
EKSPLORASI JENIS *CYBERBULLYING*

WORD CLOUD DAN HORIZONTAL BAR CHART





5 TOP KEYWORDS





RELIGION BULLYING

Focuses on religion-based hate speech. Dominated by religious identity terms (*muslim, christian, jew*) and associated stigmas (*terrorism, radical*).

EKSPLORASI JENIS *CYBERBULLYING*

WORD CLOUD DAN HORIZONTAL BAR CHART





5 TOP KEYWORDS





OTHER BULLYING

General bullying category not belonging to specific groups. Words like (*idiot, fucking, bitch, shit, stupid*) reflect high toxicity and emotional outbursts in conversations.

EKSPLORASI JENIS *CYBERBULLYING*

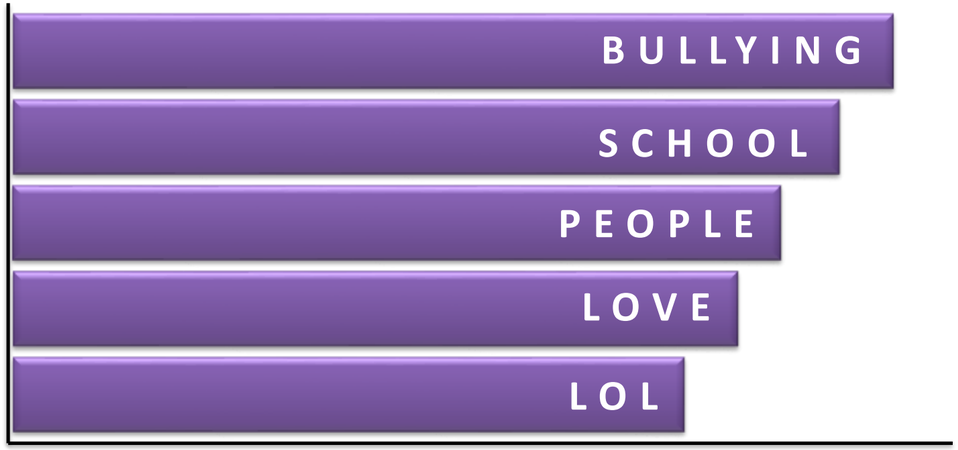
WORD CLOUD DAN HORIZONTAL BAR CHART





5 TOP KEYWORDS





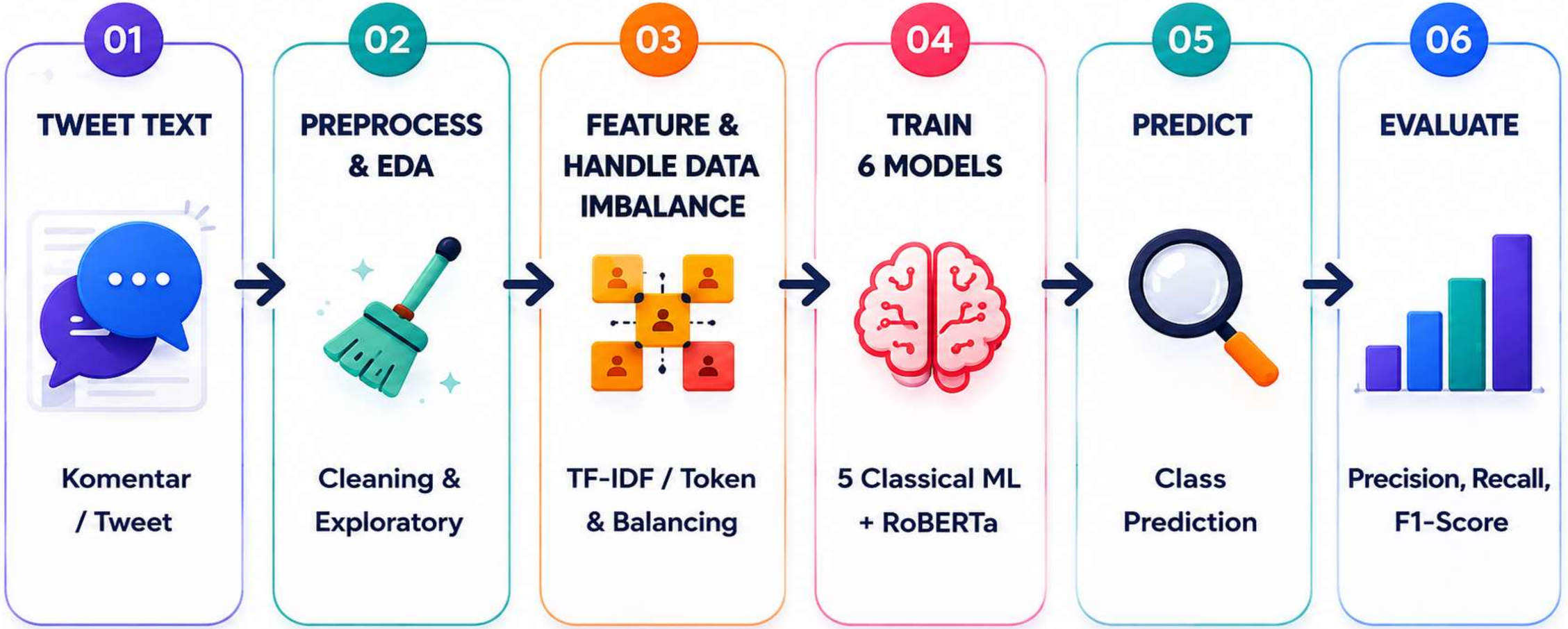
NOT BULLYING

Dominated by general discourse around school and bullying (*bullying, school, people, love, lol*). The terms are relatively neutral, containing no explicit slurs.

CLASSIC MODEL VS ROBERTA

ASPEK PERBANDINGAN	LOGISTIC REGRESSION	NAIVE BAYES	SVM (Linear)	RANDOM FOREST	XGBOOST	ROBERTA (Pre-trained (Transformer)
PRINSIP KERJA	Model linear untuk klasifikasi berbasis probabilitas	Probabilistik berbasis Teorema Bayes dengan asumsi independensi fitur	Mencari hyperplane terbaik untuk memisahkan kelas	Ensemble dari banyak pohon keputusan (boootstrapping)	Boosting ensemble yang membangun model secara bertahap	Model deep learning transformer berbasis pre-training pada teks besar
REPRESENTASI TEKS	TF-IDF, Count Vector, atau n-gram	TF-IDF, Count Vector, atau n-gram	TF-IDF, Count Vector, atau n-gram	TF-IDF, Count Vector, atau n-gram	TF-IDF, Count Vector, atau n-gram	Tokdel embedding kontekstual (subword BPE)
KEBUTUHAN DATA	Rendah – Sedang	Rendah	Sedang – Tinggi	Sedang – Tinggi	Tinggi	Tinggi (tapi bisa fine-tune pada data lebih kecil)
PERFORMA (UMUM)	★★½☆☆	★★★☆☆	★★★½☆	★★★★☆	★★★★½	★★★★★
KEKUATAN UTAMA	Sederhana, cepat, mudah diinterpretasi	Sangat cepat, efektif untuk teks pendek	Efektif pada data dimensi tinggi	Robust terhadap overfitting, bisa menangani non-linear	Performa tinggi, menangani missing value dengan baik	Memahami konteks mendalam, performa tertinggi pada NLP
KELEMAHAN UTAMA	Linear, kurang menangkap konteks kompleks	Asumsi independensi fitur sering tidak terpenuhi	Tidak menangkap konteks semantik mendalam	Butuh waktu lebih lama, kurang optimal untuk teks sangat besar	Perlu tuning hyperparameter, kompleksitas tinggi	Komputasi tinggi, perlu GPU, waktu training lebih lama
COCOK UNTUK	Baseline model, data kecil	Teks pendek, klasifikasi cepat	Teks berdimensi tinggi, akurasi baik	Data beragam, fitur banyak	Kompetisi, akurasi tinggi	NLP kompleks, akurasi terbaik

MACHINE LEARNING PIPELINE



DATA TRANSFORMATION

CLEANED DATA VOLUME

45.472

TWEETS

For *RoBERTa*

CLEANED DATA VOLUME

44.644

TWEETS

For Classic Model

AFTER PREPROCESSING STEP

- DUPLICATES HANDLING
- MISSING VALUE HANDLING
- OUTLIERS HANDLING
- CASE FOLDING HANDLING
- STOPWORDS STEMMING / LEMMATIZATION



MODELING RESULTS

CLASSIC MODEL VS RoBERTa

2 STAGES CLASSIFICATION

STAGE 1 : CYBERBULLYING VS NOT CYBERBULLYING

STAGE 2 : AGE VS ETHNICITY VS GENDER VS RELIGION VS OTHER CYBERBULLYING

CLASSIC MODEL VS ROBERTA

BEST MODEL :

RoBERTA

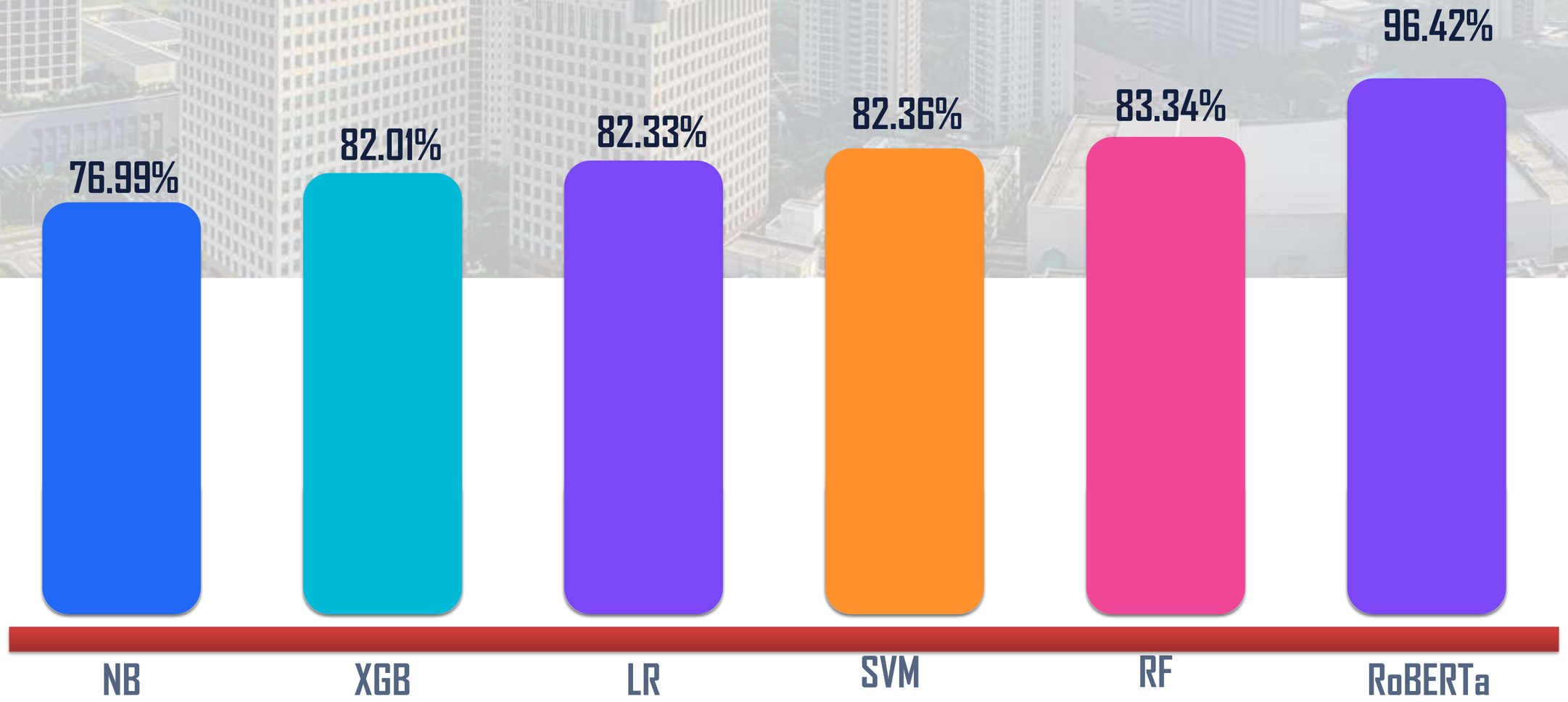
Precision (96.43%),
Recall (96.42%),
dan F1-Score (96.42%).

Achieved a 12–13% higher F1-Score compared to top classical models (Random Forest / XGBoost), proving RoBERTa's superior capability in capturing context and textual nuances.

Model	Precision	Recall	F1-Score
Logistic Regression	*83.90%*	*82.30%*	82.33%
Naive Bayes	*78.33%*	*77.75%*	76.99%
SVM	*83.33%*	*82.13%*	82.26%
Random Forest	*84.20%*	*83.18%*	83.34%
XGBoost	*85.48%*	*82.46%*	82.01%
RoBERTa	*96.43%*	*96.42%*	96.42%

F1 SCORE CAMPARISON

CLASSIC MODEL VS RoBERTa



MODEL DEPLOMENT

BEST MODEL (RoBERTa) VS BEST CLASSIC MODEL (RANDOM FOREST)

RoBERTa

	Text Tweet	Predicted Category
0	i was constantly bullied in high school by the popular kids every day	age
1	middle school bullies are literally the most cruel and evil people	age
2	getting bullied at school destroyed my self esteem when i was young	age
3	black people are useless and ruining this whole country go away	ethnicity
4	white people have no culture and ruin every neighborhood they join	ethnicity
5	Go back to your own country you dirty illegal immigrant	ethnicity
6	women are dumb bitches and should never be allowed in politics	gender
7	he is such a weak pathetic man acting like a dramatic emotional girl	gender
8	dumb blonde girls shouldn't be allowed to work in engineering	gender
9	muslims are all radical terrorists who destroy peace everywhere	religion
10	jewish people control everything and ruin the world economy	religion
11	christian fanatics are pushing their stupid brainwashing cult on us	religion
12	shut up you pathetic loser stop crying online like a whiny baby	other_cyberbullying
13	you are so ugly and fake nobody will ever like you delete your account	other_cyberbullying
14	stop posting garbage content you are completely useless and talentless	other_cyberbullying
15	had a great time at the park today with my friends and family	not_cyberbullying
16	i deeply respect older people and their incredible life wisdom	not_cyberbullying
17	Your recent video was really helpful and well explained, keep it up!	not_cyberbullying
18	happy birthday to my little brother hope you have an awesome day	not_cyberbullying
19	agree to disagree on this political topic but thanks for the discussion	not_cyberbullying

RANDOM FOREST

	Tweet Input	Hasil Prediksi Pipeline
0	i was constantly bullied in high school by the popular kids every day	age
1	middle school bullies are literally the most cruel and evil people	age
2	getting bullied at school destroyed my self esteem when i was young	age
3	black people are useless and ruining this whole country go away	ethnicity
4	white people have no culture and ruin every neighborhood they join	ethnicity
5	Go back to your own country you dirty illegal immigrant	other_cyberbullying
6	women are dumb bitches and should never be allowed in politics	ethnicity
7	he is such a weak pathetic man acting like a dramatic emotional girl	gender
8	dumb blonde girls shouldn't be allowed to work in engineering	ethnicity
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CONCLUSIONS

FINAL PROJECT

EXPLORATORY DATA ANALYSIS (EDA)

1. **The cyberbullying** dataset is divided into several main categories: Gender/Sexuality-based harassment, Religion, Age, Ethnicity/Race, and general categories (*Other Cyberbullying* & *Not Cyberbullying*).

2. **Hate speech patterns** vary across each category:
   - Gender and Sexuality: Predominantly features harassment that is often disguised as humor ("jokes").
   - Ethnicity and Religion: Employs direct insults, stereotyping, and extreme racial or hate slurs.
   - Age: Centers on school or teenage bullying environments, with terms predominantly referencing education levels.

CONCLUSIONS

FINAL PROJECT

PREPROCESSING EFFECTIVENESS & MODEL ARCHITECTURE

1. Classical Models vs Deep Learning: Model architecture selection directly impacts performance.
Classical ML Models (*Logistic Regression, Naive Bayes, SVM, Random Forest, XGBoost*) are limited by word frequency (*Bag-of-Words/TF-IDF*), capping F1-Scores at **82% – 83%** (top model: **Random Forest at 83.34%**). **RoBERTa** significantly outperforms them, achieving an **F1-Score of 96.42%** (Precision: 96.43%, Recall: 96.42%).

2. The Importance of Context: RoBERTa's strong performance confirms that cyberbullying detection requires **contextual understanding and sentence structure** (via *Self-Attention*), beyond simple keyword or slur detection.

CONCLUSIONS

FINAL PROJECT

IMPLEMENTED RECOMMENDATION

1. Best Model for Production: RoBERTa is recommended when high accuracy in detecting complex and subtle cyberbullying is the top priority.

2. Alternative Models (Resource-Constrained): If limited by computational infrastructure (no GPU / need low-latency inference), **Random Forest** or **SVM** provide the most stable classical alternatives.

CRITICAL EVALUATION

(LIMITATIONS & TECHNICAL DRAWBACKS)

FINAL PROJECT

1. High Resource Demands of RoBERTa:

- Despite its exceptional F1-Score (96.42%), RoBERTa demands heavy GPU resources and slower inference times compared to classical models during real-time deployment.

2. Risk of Bias on Sensitive Keywords:

- Identity terms (*gay*, *muslim*, *black*) show high frequency in bullying classes, risking overfitting or false positives (flagging neutral sentences containing these words as bullying).
- The dominance of the word *school* in the age category introduces potential bias due to dataset limitations.

NEXT STEPS

(FUTURE WORK & IMPROVEMENTS)

FINAL PROJECT

1. Bias Mitigation & Data Augmentation:

- Evaluate model bias by testing neutral sentences with sensitive keywords (*gay*, *muslim*) to prevent false positive bullying predictions.
- Utilize **Data Augmentation** (e.g., *back-translation*) to address underrepresented or inconsistent data classes.

2. Deployment Optimization (Model Compression):

- For real-time application, implement **quantization** or adopt lighter Transformer models (**DistilRoBERTa**, **MobileBERT**) to trim model size and boost inference speed with minimal loss in accuracy.



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THANK YOU

*"Machine Learning: 80% cleaning data,
20% complaining about cleaning data."*